Radiance

Wireless 65% keyboard with German ISO layout, two rotary encoders and an OLED display. User guide and firmware reference.



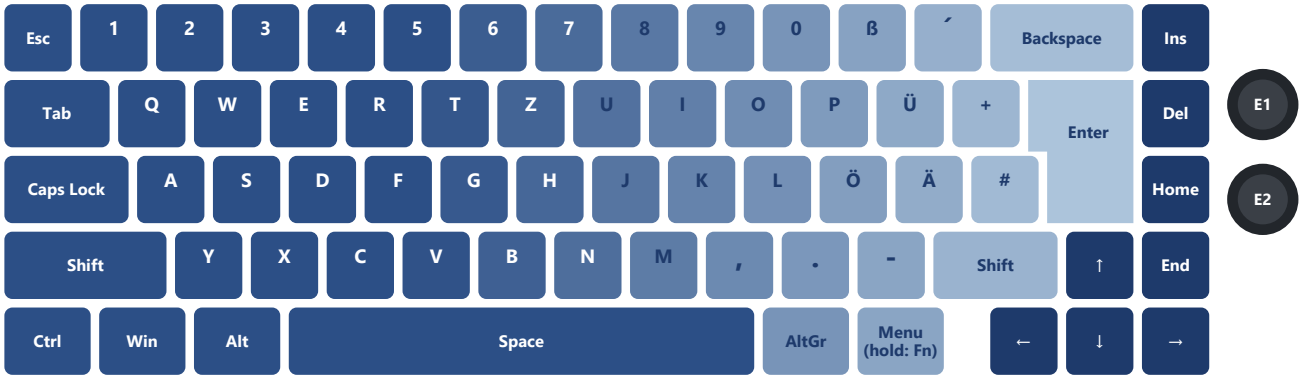
At a glance

Layout	67 keys, German ISO (QWERTZ), 65% with arrow cluster and Ins / Del / Home / End column
Controller	Pro Micro-compatible nRF52840 board (firmware target **nice_nano_v2**)
Connection	USB-C or Bluetooth Low Energy, up to 5 paired devices
Firmware	ZMK v0.3.0 with ZMK Studio (live remapping over USB), NKRO
Extras	Two EC11 rotary encoders with push buttons, 128×32 SSD1306 OLED
Firmware repo	github.com/jonasradke/zmk-config-radiance

Keymap

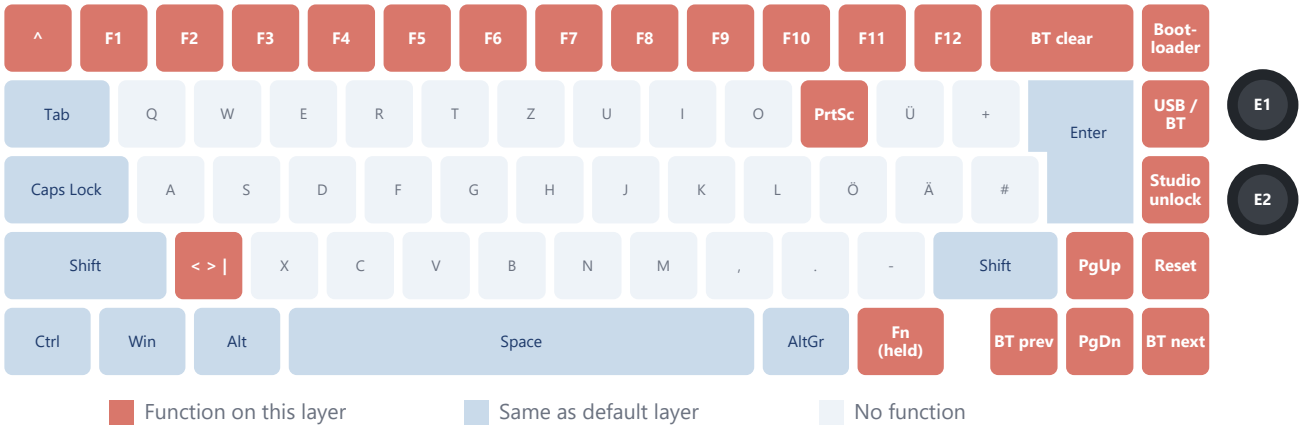
The keyboard has two layers. The **Default** layer is what you type on. The **Function** layer is active while you hold the **Menu** key (bottom row, right of AltGr). A short tap on Menu still sends the normal context-menu key.

Default layer



Set the operating system's keyboard layout to **German**. The keyboard sends key positions, and the OS turns them into the printed characters (ß, Ü, Ö, Ä, # and so on). AltGr works as usual: AltGr+Q = @, AltGr+E = €, AltGr+7/8/9/0 = { [ ] }, AltGr++ = ~.

Function layer (hold Menu)



Keys	Action
Fn + Esc	^ (the board has no key left of 1)
Fn + 1 … 0, ß, ´	F1 … F12
Fn + Y	< · with Shift: > · with AltGr: | (the board has no <> key)
Fn + P	Print Screen
Fn + ↑ / ↓	Page Up / Page Down
Fn + ← / →	Previous / next Bluetooth profile
Fn + Backspace	Clear the pairing of the current Bluetooth profile
Fn + Del	Toggle output between USB and Bluetooth
Fn + Home	Unlock ZMK Studio
Fn + End	Restart the keyboard
Fn + Ins	Enter the bootloader (for flashing firmware)

Rotary encoders

Both encoders behave the same on every layer.

Encoder	Turn	Press
Encoder 1	Page Down / Page Up	Play / Pause
Encoder 2	Volume up / Volume down	Mute

The turn direction depends on how the encoder's A/B pins are wired. If it runs the wrong way, swap the two keycodes in the `sensor-bindings` line of the keymap.

Connecting

USB

Plug in a USB-C cable. While USB is connected, the keyboard types over USB and charges its battery. To type over Bluetooth while the cable is plugged in (for example to charge), press **Fn + Del**.

Bluetooth

The keyboard stores up to five Bluetooth profiles, one device each.

Step	What to do
1. Pick a profile	Press Fn + ← / → until the display shows the profile you want.
2. Free it	If the profile was used before, press Fn + Backspace to clear it.
3. Pair	Open Bluetooth settings on the computer and select **Radiance**.
Switch devices	Fn + ← / → jumps between paired devices.

When re-pairing, remove **Radiance** from the computer's device list too, otherwise the two sides keep different keys and the connection fails.

Display

The OLED shows the active output (USB or Bluetooth profile), battery level in percent, the current layer name ("Default" or "Function") and typing speed (WPM).

Troubleshooting

Problem	Fix
Letters come out wrong (Y and Z swapped, ; instead of ö)	The computer uses an English layout. Switch it to German.
One key does nothing	Its switch may be on the other of two matrix positions, or the solder joint is bad. Look it up in the matrix diagram and check the binding and the joint.
Bluetooth won't reconnect	Fn + Backspace on that profile, remove Radiance from the computer, pair again.
Types on the wrong device	Fn + Del to switch USB/Bluetooth, Fn + ← / → to switch profile.
Studio won't connect	Use USB (not Bluetooth), a Chromium browser or the desktop app, and press Fn + Home.
Display stays dark	Check the OLED's four pins. The display turns off when the keyboard is idle.

Changing the keymap

With ZMK Studio (no flashing)

1. Connect the keyboard over USB.
2. Open **zmk.studio** in Chrome or Edge, or the ZMK Studio desktop app, and select Radiance.
3. When Studio asks, press **Fn + Home** to unlock it.
4. Click a key and pick a new behavior. Changes are saved on the keyboard immediately.

Studio changes override the keymap file. To go back to the keymap from the repository, use Restore stock settings in Studio.

By editing the firmware

The keymap lives in the GitHub repository. Every push builds new firmware with GitHub Actions.

File	Contents
config/boards/shields/radiance/radiance.keymap	Key bindings for both layers and the encoders
config/boards/shields/radiance/radiance.overlay	Pins, matrix, encoders, display
config/boards/shields/radiance/radiance-layouts.dtsi	Key positions and sizes shown in ZMK Studio
config/radiance.conf	Feature switches (display widgets, NKRO, Bluetooth power)
build.yaml	Build target: nice_nano_v2 + radiance shield, Studio enabled

Binding rows in the keymap follow the matrix (16, 16, 16, 15, 11 entries). Some keys appear twice because the PCB has two possible switch positions for them; give both the same binding.

Flashing firmware

1. On GitHub, open **Actions**, pick the latest successful run and download the **firmware** artifact. Unzip it to get the `.uf2` file.
2. Connect the keyboard over USB and press **Fn + Ins** (or double-tap the reset button on the controller). A USB drive appears (usually named **NICENANO**).
3. Copy the `.uf2` file onto that drive. The keyboard restarts with the new firmware when the copy is done.

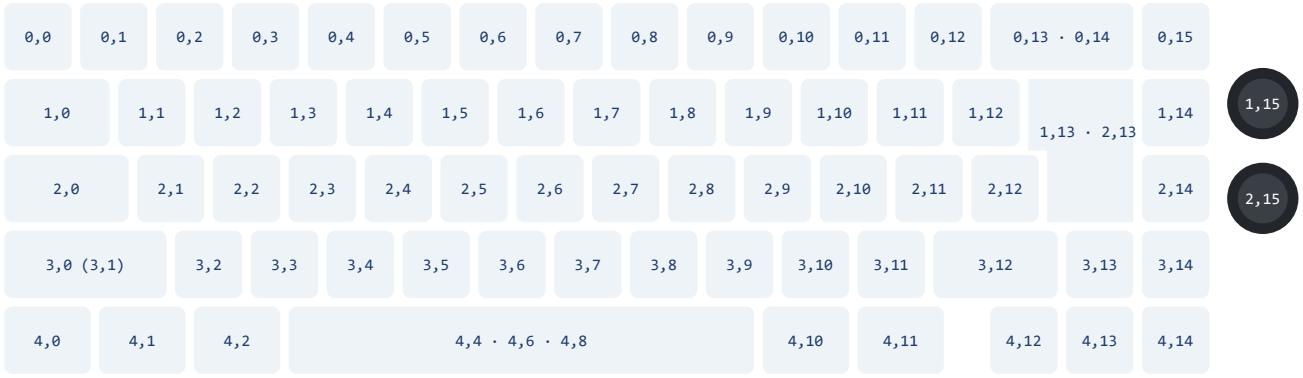
Recovering from a bad flash

If the keyboard stops responding, double-tap the reset button on the controller to reach the bootloader and flash a known-good `.uf2`. To wipe all stored settings (Bluetooth pairings, Studio changes), flash the `settings_reset` firmware for nice_nano_v2 from the ZMK project, then flash Radiance again.

Hardware reference

Matrix positions

Each key's row,column in the matrix. Useful when a key does nothing: it tells you which binding in the keymap belongs to it. Keys with two positions have two possible switch footprints on the PCB.



Wiring

Function	Controller pins	Notes
Matrix rows 0–4	10, 16, 14, 15, 18	Inputs, active low, internal pull-up
Column select	6, 7, 8, 9	4-bit address decoded to 16 columns by the logic ICs (gpio-demux scan)
Encoder 1	A = 4, B = 5	EC11, 80 steps, 20 triggers per rotation; push = matrix 1,15
Encoder 2	A = 21, B = 20	EC11, 80 steps, 20 triggers per rotation; push = matrix 2,15
OLED	I²C (SDA/SCL of the Pro Micro footprint)	SSD1306, 128×32, address 0x3C

Firmware settings

Setting	Value
Keyboard name	Radiance
HID report	NKRO (all keys can be pressed at once)
Bluetooth transmit power	+8 dBm
Display widgets	Output status, battery percentage, layer, WPM
Matrix scan	Every 1 ms